WEEK 3 – PROCESS SCHEDULING SIMULATOR

Introduction

Process scheduling is an operating system function that decides which process gets CPU time. This project implements six scheduling algorithms in C and provides a menu-driven interface to test their behaviour.

Objectives

- Implement CPU scheduling algorithms in C.
- Accept arrival time, burst time and priority.
- Understand preemptive and non-preemptive scheduling.
- Evaluate scheduling performance using process metrics.
- Test algorithms using different datasets.

Tools Used

- **Language:** C
- **Compiler:** GCC
- **Environment:** Ubuntu Linux / WSL
- **Interface:** Command-Line Interface (CLI)
- **Version Control:** GitHub

Algorithms Implemented

1. **FCFS:** Executes processes in arrival order.
2. **SJF Non-Preemptive:** Selects the available process with the shortest burst time.
3. **SJF Preemptive:** Selects the process with the shortest remaining time.
4. **Priority Non-Preemptive:** Executes the highest-priority available process without interruption.
5. **Priority Preemptive:** Allows a higher-priority process to interrupt the running process.
6. **Round Robin:** Gives each ready process a fixed time quantum in turn.

Performance Metrics

- **Turnaround Time (TAT) = Completion Time – Arrival Time**
- **Waiting Time (WT) = Turnaround Time – Burst Time**

- **Response Time (RT) = First Start Time − Arrival Time**
- **Average WT = Total WT / Number of Processes**
- **Average TAT = Total TAT / Number of Processes**
- **CPU Utilization (%) = (CPU Busy Time / Total Elapsed Time) × 100**
-

Test Cases

Test Case 1

Process	Arrival	Burst	Priority
P1	0	5	2
P2	1	3	1
P3	2	4	3

Round Robin quantum: **2**

```
   PROCESS SCHEDULING SIMULATOR
====================================
Enter number of processes (1-100): 3

Process P1
Arrival Time: 0
Burst Time: 5
Priority (smaller number = higher priority): 2

Process P2
Arrival Time: 1
Burst Time: 3
Priority (smaller number = higher priority): 1

Process P3
Arrival Time: 2
Burst Time: 4
Priority (smaller number = higher priority): 3

========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 1

FCFS Gantt Chart:
| P1 | P2 | P3 |
Final time: 12

PID     AT      BT      CT      TAT     WT      RT
P1      0       5       5       5       0       0
P2      1       3       8       7       4       4
P3      2       4       12      10      6       6
Average Waiting Time: 3.33
Average Turnaround Time: 7.33
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 2

SJF Non-Preemptive Gantt Chart:
| P1 | P2 | P3 |
Final time: 12

PID     AT      BT      CT      TAT     WT      RT
P1      0       5       5       5       0       0
P2      1       3       8       7       4       4
P3      2       4       12      10      6       6
Average Waiting Time: 3.33
Average Turnaround Time: 7.33
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 3

SJF Preemptive (Shortest Remaining Time First)

Gantt Chart:
| P1 | P2 | P2 | P2 | P1 | P1 | P1 | P1 | P3 | P3 | P3 | P3 |
Final time: 12

PID     AT      BT      CT      TAT     WT      RT
P1      0       5       8       8       3       0
P2      1       3       4       3       0       0
P3      2       4       12      10      6       6
Average Waiting Time: 3.00
Average Turnaround Time: 7.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 4

Priority Non-Preemptive Gantt Chart:
| P1 | P2 | P3 |
Final time: 12

PID     AT      BT      CT      TAT     WT      RT
P1      0       5       5       5       0       0
P2      1       3       8       7       4       4
P3      2       4       12      10      6       6
Average Waiting Time: 3.33
Average Turnaround Time: 7.33
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 5

Priority Preemptive

Gantt Chart:
| P1 | P2 | P2 | P2 | P1 | P1 | P1 | P1 | P3 | P3 | P3 | P3 |
Final time: 12

PID     AT      BT      CT      TAT     WT      RT
P1      0       5       8       8       3       0
P2      1       3       4       3       0       0
P3      2       4       12      10      6       6
Average Waiting Time: 3.00
Average Turnaround Time: 7.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 6
Enter time quantum: 2

Round Robin Gantt Chart (Quantum = 2):
| P1 | P2 | P3 | P1 | P2 | P3 | P1 |
Final time: 12

PID     AT      BT      CT      TAT     WT      RT
P1      0       5       12      12      7       0
P2      1       3       9       8       5       1
P3      2       4       11      9       5       2
Average Waiting Time: 5.67
Average Turnaround Time: 9.67
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 7
Exiting simulator.
```

Test Case 2

Process	Arrival	Burst	Priority
P1	0	7	3
P2	2	4	1
P3	4	2	2

Round Robin quantum: **3**

```
====================================
   PROCESS SCHEDULING SIMULATOR
====================================
Enter number of processes (1-100): 3

Process P1
Arrival Time: 0
Burst Time: 7
Priority (smaller number = higher priority): 3

Process P2
Arrival Time: 2
Burst Time: 4
Priority (smaller number = higher priority): 1

Process P3
Arrival Time: 4
Burst Time: 2
Priority (smaller number = higher priority): 2

========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 1

FCFS Gantt Chart:
| P1 | P2 | P3 |
Final time: 13

PID     AT      BT      CT      TAT     WT      RT
P1      0       7       7       7       0       0
P2      2       4       11      9       5       5
P3      4       2       13      9       7       7
Average Waiting Time: 4.00
Average Turnaround Time: 8.33
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 2

SJF Non-Preemptive Gantt Chart:
| P1 | P3 | P2 |
Final time: 13

PID     AT      BT      CT      TAT     WT      RT
P1      0       7       7       7       0       0
P2      2       4       13      11      7       7
P3      4       2       9       5       3       3
Average Waiting Time: 3.33
Average Turnaround Time: 7.67
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 3

SJF Preemptive (Shortest Remaining Time First)

Gantt Chart:
| P1 | P1 | P2 | P2 | P2 | P2 | P3 | P3 | P1 | P1 | P1 | P1 | P1 |
Final time: 13

PID     AT      BT      CT      TAT     WT      RT
P1      0       7       13      13      6       0
P2      2       4       6       4       0       0
P3      4       2       8       4       2       2
Average Waiting Time: 2.67
Average Turnaround Time: 7.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 4

Priority Non-Preemptive Gantt Chart:
| P1 | P2 | P3 |
Final time: 13

PID     AT      BT      CT      TAT     WT      RT
P1      0       7       7       7       0       0
P2      2       4       11      9       5       5
P3      4       2       13      9       7       7
Average Waiting Time: 4.00
Average Turnaround Time: 8.33
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 5

Priority Preemptive

Gantt Chart:
| P1 | P1 | P2 | P2 | P2 | P2 | P3 | P3 | P1 | P1 | P1 | P1 | P1 |
Final time: 13

PID     AT      BT      CT      TAT     WT      RT
P1      0       7       13      13      6       0
P2      2       4       6       4       0       0
P3      4       2       8       4       2       2
Average Waiting Time: 2.67
Average Turnaround Time: 7.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 6
Enter time quantum: 3

Round Robin Gantt Chart (Quantum = 3):
| P1 | P2 | P1 | P3 | P2 | P1 |
Final time: 13

PID     AT      BT      CT      TAT     WT      RT
P1      0       7       13      13      6       0
P2      2       4       12      10      6       1
P3      4       2       11      7       5       5
Average Waiting Time: 5.67
Average Turnaround Time: 10.00
CPU Utilization: 100.00%
```

Test Case 3

Process	Arrival	Burst	Priority
P1	1	8	2
P2	0	3	1
P3	2	5	3

Round Robin quantum: **2**

```
==================================
   PROCESS SCHEDULING SIMULATOR
==================================
Enter number of processes (1-100): 3

Process P1
Arrival Time: 1
Burst Time: 8
Priority (smaller number = higher priority): 2

Process P2
Arrival Time: 0
Burst Time: 3
Priority (smaller number = higher priority): 1

Process P3
Arrival Time: 2
Burst Time: 5
Priority (smaller number = higher priority): 3

========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 1

FCFS Gantt Chart:
| P2 | P1 | P3 |
Final time: 16

PID     AT      BT      CT      TAT     WT      RT
P1      1       8       11      10      2       2
P2      0       3       3       3       0       0
P3      2       5       16      14      9       9
Average Waiting Time: 3.67
Average Turnaround Time: 9.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 2

SJF Non-Preemptive Gantt Chart:
| P2 | P3 | P1 |
Final time: 16

PID     AT      BT      CT      TAT     WT      RT
P1      1       8       16      15      7       7
P2      0       3       3       3       0       0
P3      2       5       8       6       1       1
Average Waiting Time: 2.67
Average Turnaround Time: 8.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 3

SJF Preemptive (Shortest Remaining Time First)

Gantt Chart:
| P2 | P2 | P2 | P3 | P3 | P3 | P3 | P3 | P1 | P1 | P1 | P1 | P1 | P1 | P1 | P1 |
Final time: 16

PID     AT      BT      CT      TAT     WT      RT
P1      1       8       16      15      7       7
P2      0       3       3       3       0       0
P3      2       5       8       6       1       1
Average Waiting Time: 2.67
Average Turnaround Time: 8.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 4

Priority Non-Preemptive Gantt Chart:
| P2 | P1 | P3 |
Final time: 16

PID     AT      BT      CT      TAT     WT      RT
P1      1       8       11      10      2       2
P2      0       3       3       3       0       0
P3      2       5       16      14      9       9
Average Waiting Time: 3.67
Average Turnaround Time: 9.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 5

Priority Preemptive

Gantt Chart:
| P2 | P2 | P2 | P1 | P1 | P1 | P1 | P1 | P1 | P1 | P1 | P3 | P3 | P3 | P3 | P3 |
Final time: 16

PID     AT      BT      CT      TAT     WT      RT
P1      1       8       11      10      2       2
P2      0       3       3       3       0       0
P3      2       5       16      14      9       9
Average Waiting Time: 3.67
Average Turnaround Time: 9.00
CPU Utilization: 100.00%
```

```
========== SCHEDULING MENU ==========
1. FCFS
2. SJF (Non-Preemptive)
3. SJF (Preemptive)
4. Priority (Non-Preemptive)
5. Priority (Preemptive)
6. Round Robin
7. Exit
Enter your choice: 6
Enter time quantum: 2

Round Robin Gantt Chart (Quantum = 2):
| P2 | P1 | P3 | P2 | P1 | P3 | P1 | P3 | P1 |
Final time: 16

PID     AT      BT      CT      TAT     WT      RT
P1      1       8       16      15      7       1
P2      0       3       7       7       4       0
P3      2       5       14      12      7       2
Average Waiting Time: 6.00
Average Turnaround Time: 11.33
CPU Utilization: 100.00%
```

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7. Performance Comparison

Test Case 1

Scheduling Algorithm	**Average Waiting Time**	**Average Turnaround Time**	**CPU Utilization (%)**
FCFS	3.33	7.33	100.00
SJF Non-Preemptive	3.33	7.33	100.00
SJF Preemptive	3.00	7.00	100.00
Priority Non-Preemptive	3.33	7.33	100.00
Priority Preemptive	3.00	7.00	100.00
Round Robin (Q = 2)	5.67	9.67	100.00

Test Case 2

Scheduling Algorithm	Average Waiting Time	Average Turnaround Time	CPU Utilization (%)
FCFS	4.00	8.33	100.00
SJF Non-Preemptive	3.33	7.67	100.00
SJF Preemptive	2.67	7.00	100.00
Priority Non-Preemptive	4.00	8.33	100.00
Priority Preemptive	2.67	7.00	100.00
Round Robin (Q = 3)	5.67	10.00	100.00

Test Case 3

Scheduling Algorithm	Average Waiting Time	Average Turnaround Time	CPU Utilization (%)
FCFS	3.67	9.00	100.00
SJF Non-Preemptive	2.67	8.00	100.00
SJF Preemptive	2.67	8.00	100.00
Priority Non-Preemptive	3.67	9.00	100.00
Priority Preemptive	3.67	9.00	100.00
Round Robin (Q = 2)	6.00	11.33	100.00

8. Conclusion

The Process Scheduling Simulator demonstrates six CPU scheduling algorithms using C. It allows users to enter process details and select an algorithm through a menu. Testing with different datasets helps explain how scheduling methods affect process execution and performance.